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$$f(x) = \ln\left(\tan\left(\frac{x}{2}\right)\right)$$

$$f'(x) = \frac{1}{\tan\left(\frac{x}{2}\right)} \cdot \left(\tan\left(\frac{x}{2}\right)\right)' = \frac{\sec^2\left(\frac{x}{2}\right)\left(\frac{x}{2}\right)'}{\tan\left(\frac{x}{2}\right)}$$

$$= \frac{1}{\tan\left(\frac{x}{2}\right)} \cdot \sec^2\left(\frac{x}{2}\right) \cdot \frac{1}{2}$$

$$= \frac{\sec^2\left(\frac{x}{2}\right)}{2\tan\left(\frac{x}{2}\right)}$$

$$= \frac{1}{2\cos\left(\frac{x}{2}\right)\sin\left(\frac{x}{2}\right)}$$

$$= \frac{1}{\sin x}$$

References